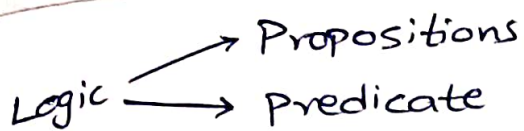


Logics And Proofs



- Logic is important for both program design and mathematical reasoning.
- The propositional logic is a system which is based on propositions where each proposition has syntax and symantic.
- Syntax represents the ways to represent a statement. Symantic represents the meaning i.e. the truth value (T/F).

→ Proposition / Statement

A proposition is a declarative sentence that describes a fact where it can be either true/false but not both.

A proposition is always represented by lower case letters such as $p, q, r, s, \dots$

ex 1) P : Bangalore is the capital of Karnataka.

- It is a proposition / Primitive statement.
- It's truth value is True.

2) q : $112 < 102$

- It is a proposition / Primitive statement.
- It's truth value is False.

3) s : $x+2 > 5$

- It is not a proposition coz the truthness of the statement depends on the value of 'x' which is unknown.

Such kind of statement is referred to as "Open Statement".

4) t : What a beautiful day it is !

- It is not a proposition coz it is an exclamatory sentence.

NOTE:- Exclamatory Sentences and Commanding Sentences are not propositions as their truth value cannot be determined.

Compound Statements:

They are made up of 2 (or) more propositions connected by logical connectives.

Logical connectives are:

- 1) Negation (NOT, $\neg$)
- 2) Conjunction (AND, $\wedge$)
- 3) Disjunction (OR, $\vee$)
- 4) Implication (if, then, $\rightarrow$)
- 5) Biconditional (if and only if, $\leftrightarrow$)

ex: P: Mangalore is a beautiful city

q: Bangalore is the capital of Karnataka.

1) $P \wedge q$: Mangalore is a beautiful city and Bangalore is capital of Karnataka.

2) $P \vee q$: Mangalore is a beautiful city or Bangalore is capital of Karnataka.

Truth tables for Connectivity

1) Negation:- Negation of the given proposition 'P' is denoted by $\neg P$.

If truth value of P is 1 then its negation is 0 and vice-versa.

P	$\neg P$
0	1
1	0

2) Conjunction:- Compound Proposition obtained by combining two proposition with the word 'AND' is called conjunction is denoted by $P \wedge q$ if P, q are propositions.

Thumb rule: If P, q are true then $P \wedge q$ is true

P	q	$P \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

3) Disjunction:- Compound proposition obtained by combining two proposition with the word 'OR' is called Disjunction is denoted by $P \vee Q$.

Thumb Rule:- $P \vee Q$ is false only when P & Q are false.

P	Q	$P \vee Q$
0	0	0
0	1	1
1	0	1
1	1	1

4) Implication:- Compound proposition obtained by combining 2 propositions with the word "if, then" is called implication. denoted by $P \rightarrow Q$.

Thumb Rule:- $P \rightarrow Q$ is false only when P is true and Q is false.

P	Q	$P \rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

5) Biconditional:- Compound propositions obtained by combining the propositions with the words XNOR 'if and only if' is called Biconditional represented as $P \leftrightarrow Q$.

Thumb rule:- $P \leftrightarrow Q$ is true only when both P and Q have same truth values.

P	Q	$P \leftrightarrow Q$
0	0	1
0	1	0
1	0	0
1	1	1

NOTE:- $\oplus$ denotes the connectivity XOR and the Thumb rule is $P \oplus Q$ is false when both P & Q have same truth values.

NOTE:- We need to construct the truth table by considering all possible combinations of primitive statements i.e. if there are 'n' propositions then the truth table contains 2^n combinations.

Q) Construct the truth tables for the below compound propositions.

i) $(p \vee q) \wedge r$

P	q	r	$p \vee q$	$(p \vee q) \wedge r$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	0
1	1	1	1	1

ii) $q \leftrightarrow (\neg p \vee \neg q)$

P	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$q \leftrightarrow (\neg p \vee \neg q)$
0	0	1	1	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	1	0	0	0	0

iii) $(p \wedge q) \rightarrow \neg r$

P	q	r	$p \wedge q$	$\neg r$	$p \wedge q \rightarrow \neg r$
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	0	1
1	0	0	0	1	1
1	0	1	0	0	1
1	1	0	1	1	1
1	1	1	1	0	0

Q) Let p, q be primitive statements for which $p \leftrightarrow q$ is false. Determine the truth value of the following compound proposition with appropriate justification.

a) $p \wedge q$ b) $\neg p \vee q$ c) $q \rightarrow p$ d) $\neg q \rightarrow \neg p$

Soln:- $p \rightarrow q \Rightarrow \text{False}$

① ②

we know that since the assumption is $p \rightarrow q$ is false, we analyze that the truth value of p is true and truth value of q is false. Therefore, truth values are $p=1, q=0$.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$
1	0	0	1	0	0	1	0

Q) Find the truth values of p, q, r, s, t for the compound proposition. $((p \wedge q) \wedge r) \rightarrow (s \vee t)$ is false.

$$p=1, q=1, r=1, s=0, t=0$$

NOTE:- LHS of implication is called as premise to that RHS of implication is called as conclusion $p \rightarrow q$ here, p is premise, q is conclusion.

1) Tautology:- When all the truth values are 1 (True) for all possible combinations then the compound proposition is tautology.

2) Contradiction:- When all the truth values are 0 (False) for all possible combinations then the compound proposition is Contradiction.

3) Contingency:- When all the truth values are Neither 0 nor 1 for all combinations then the compound Proposition is Contingency.

Q) verify that

i) $(p \vee q) \vee (p \rightarrow q)$ is a tautology.

Sol:-

p	q	$p \vee q$	$p \rightarrow q$	$(p \vee q) \vee (p \rightarrow q)$
0	0	0	1	1
0	1	1	1	1
1	0	1	0	1
1	1	1	1	1

For all combination truth table is 1 i.e true so the given proposition is a tautology.

ii) $[(P \rightarrow (Q \rightarrow R))] \rightarrow [(P \rightarrow Q) \rightarrow (P \rightarrow R)]$ is a tautology.

P	Q	R	$P \rightarrow R$	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow (Q \rightarrow R)$	$(P \rightarrow Q) \rightarrow (P \rightarrow R)$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	1
1	0	1	1	0	1	1	1
1	1	0	0	1	0	0	0
1	1	1	1	1	1	1	1

$[(P \rightarrow (Q \rightarrow R))] \rightarrow [(P \rightarrow Q) \rightarrow (P \rightarrow R)]$
1
1
1
1
1
1
0
1

iii) $(P \nleftrightarrow Q) \wedge (P \leftrightarrow Q)$ is contradiction.

P	Q	$P \nleftrightarrow Q$	$P \leftrightarrow Q$	$(P \nleftrightarrow Q) \wedge (P \leftrightarrow Q)$
0	0	0	1	0
0	1	1	0	0
1	0	1	0	0
1	1	0	1	0

→ Logical Equivalence

Two propositions U & V are said to be logically equivalent whenever U & V have same truth values and represented as $U \leftrightarrow V$ (or) $U \equiv V$ in equivalently we can consider $U \leftrightarrow V$ is tautology.

Two ways to prove that the given propositions are logically equivalent are:-

1) Truth Tables

2) Laws of Logic

* Problems on Truth Tables proving logical equivalence:-

1) $P \rightarrow Q \equiv \neg P \vee Q$

To prove that the above logical equivalence, we need to construct the truth table and verify whether the truth values of $P \rightarrow Q$ and $\neg P \vee Q$ are the same.

We can now observe that column 1 & 2 do have same truth values. Hence, we can conclude that the given propositions are logically equivalent.

P	Q	① $P \rightarrow Q$	② $\neg P \vee Q$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

2) Show that the given propositions are logically equivalent using truth table:

(i) $P \vee (P \wedge Q) \Leftrightarrow P$

Let $f = P \vee (P \wedge Q) \Leftrightarrow P$

P	Q	$P \wedge Q$	$P \vee (P \wedge Q)$	f
0	0	0	0	✓
0	1	0	0	✓
1	0	0	1	✓
1	1	1	1	✓

(ii) $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$

Let $g = P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$

P	Q	R	$P \wedge Q$	$Q \rightarrow R$	$P \rightarrow (Q \rightarrow R)$	$(P \wedge Q) \rightarrow R$
0	0	0	0	1	1	1
0	0	1	0	1	1	1
0	1	0	0	0	1	1
0	1	1	0	1	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	0	0	0
1	1	1	1	1	1	1

NOTE:- We can replace, $P \rightarrow Q$ with $\neg P \vee Q$ when it is required in problems to solve.

Laws of Logic

Consider p, q & r are the propositions, T_0 & F_0 representing a truth value True & false respectively.

1) Commutative Law: $p \vee q \iff q \vee p$
 $p \wedge q \iff q \wedge p$

2) Associative Law: $(p \vee q) \vee r \iff p \vee (q \vee r)$
 $(p \wedge q) \wedge r \iff p \wedge (q \wedge r)$

3) Distributive Law: $p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$
 $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$

4) Identity Law: $p \wedge T_0 \iff p$
 $p \vee F_0 \iff p$

5) Inverse Law: $p \wedge \neg p \iff F_0$
 $p \vee \neg p \iff T_0$

6) De Morgan's Law: $\neg(p \wedge q) \iff \neg p \vee \neg q$
 $\neg(p \vee q) \iff \neg p \wedge \neg q$

7) Law of Double Negation: $\neg(\neg p) \iff p$
 $\neg\neg p \iff p$

8) Absorption Law: $p \vee (p \wedge q) \iff p$
 $p \wedge (p \vee q) \iff p$

9) Idempotent Law: $p \vee p \iff p$
 $p \wedge p \iff p$

10) Domination Law: $p \vee T_0 \iff T_0$
 $p \wedge F_0 \iff F_0$

Q) State and verify absorption law & Demorgan's Law.

Sol: For any two propositions p & q , Absorption Law States that $p \vee (p \wedge q) \iff p$ and $p \wedge (p \vee q) \iff p$

Us, Demorgan law states that $\neg(p \wedge q) \iff \neg p \vee \neg q$ and $\neg(p \vee q) \iff \neg p \wedge \neg q$.

1) Absorption law.

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$P \vee Q$	$P \vee (P \wedge Q)$	$P \wedge (P \vee Q)$
0	0	1	1	0	0	0	0
0	1	1	0	0	1	0	0
1	0	0	1	0	1	1	1
1	1	0	0	1	1	1	1

2) Demorgans Law

P	Q	$\neg P$	$\neg Q$	$\neg(P \wedge Q)$	$\neg(P \vee Q)$	$\neg P \vee \neg Q$	$\neg P \wedge \neg Q$
0	0	1	1	1	1	1	1
0	1	1	0	1	0	1	0
1	0	0	1	1	0	1	0
1	1	0	0	0	0	0	0

Q) show given propositions are logically equivalent using laws of logic.

i) $(P \vee Q) \wedge \neg(\neg P \wedge Q) \iff P$

Sol LHS $\iff (P \vee Q) \wedge (\neg \neg P \vee \neg Q)$

$\iff (P \vee Q) \wedge (P \vee \neg Q)$

$\iff (P \vee Q) \wedge (P \vee \neg Q) \iff P \vee (Q \wedge \neg Q)$

$\iff P \vee F_0$

$\iff P \iff \text{RHS}$

Hence proved

ii) $P \rightarrow (Q \rightarrow R) \iff (P \wedge Q) \rightarrow R$

Sol: LHS $\iff P \rightarrow (Q \rightarrow R)$

$\iff \neg P \vee (\neg Q \vee R) \iff (\neg P \vee \neg Q) \vee R$

$\iff \neg(P \wedge Q) \vee R$

$\iff (P \wedge Q) \rightarrow R \iff \text{RHS}$

iii) $(P \wedge Q) \vee (P \wedge \neg Q) \iff P$

Sol LHS $\iff (P \wedge Q) \vee (P \wedge \neg Q)$

$\iff P \wedge (Q \vee \neg Q)$

$\iff P \wedge T_0 \iff P$

$\iff \text{RHS}$

$$iv) [(p \vee q) \wedge (p \vee \neg q)] \vee r \Leftrightarrow (p \vee r)$$

$$\begin{aligned} \underline{\text{Sol}} \quad \text{LHS} &\Leftrightarrow [(p \vee q) \wedge (p \vee \neg q)] \vee r \\ &\Leftrightarrow [p \vee (q \wedge \neg q)] \vee r \Leftrightarrow [p \vee F] \vee r \\ &\Leftrightarrow p \vee r \Leftrightarrow \text{RHS} \end{aligned}$$

$$v) (p \rightarrow q) \wedge (\neg q \wedge (r \vee \neg r)) \Leftrightarrow \neg(r \vee p)$$

$$\begin{aligned} \underline{\text{Sol}} \quad \text{LHS} &\Leftrightarrow (p \rightarrow q) \wedge (\neg q \wedge (r \vee \neg r)) \\ &\Leftrightarrow (\neg p \vee q) \wedge (\neg q \wedge (r \vee \neg r)) \\ &\Leftrightarrow (\neg p \vee q) \wedge \neg q \quad \longrightarrow \text{Absorption law} \\ &\Leftrightarrow (\neg q \wedge q) \vee (\neg q \wedge \neg p) \\ &\Leftrightarrow F \vee \neg(r \vee p) \Leftrightarrow \neg(r \vee p) \\ &\Leftrightarrow \text{RHS} \end{aligned}$$

$$vi) (p \vee q) \rightarrow r \Leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$$

$$\begin{aligned} \underline{\text{Sol}} \quad \text{LHS} &\Leftrightarrow (p \vee q) \rightarrow r \\ &\Leftrightarrow \neg(p \vee q) \vee r \\ &\Leftrightarrow (\neg p \wedge \neg q) \vee r \Leftrightarrow (\neg p \vee r) \wedge (\neg q \vee r) \\ &\Leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r) \\ &\Leftrightarrow \text{RHS} \end{aligned}$$

$$vii) (\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r) \Leftrightarrow (p \wedge q)$$

$$\begin{aligned} \underline{\text{LHS}} &\Leftrightarrow (\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r) \quad \text{--- De Morgan's} \\ &\Leftrightarrow \neg(p \wedge q) \rightarrow (p \wedge q \wedge r) \\ &\Leftrightarrow \neg\neg(p \wedge q) \vee (p \wedge q \wedge r) \quad \text{--- } p \rightarrow q \Leftrightarrow \neg p \vee q \\ &\Leftrightarrow (p \wedge q) \vee (p \wedge q \wedge r) \quad \text{--- Double Negation} \\ &\Leftrightarrow p \wedge (q \vee (q \wedge r)) \quad \text{--- Distributive} \\ &\Leftrightarrow p \wedge q \Leftrightarrow \text{RHS} \quad \text{--- Absorption} \end{aligned}$$

$$ix) (p \vee q) \vee (\neg p \wedge \neg q \wedge r) \iff (p \vee q \vee r)$$

$$\text{sol: } (p \vee q) \vee (\neg p \wedge \neg q \wedge r)$$

$$\iff (p \vee q) \vee (\neg(p \vee q) \wedge r)$$

$$\iff ((p \vee q) \vee (\neg(p \vee q))) \wedge ((p \vee q) \vee r)$$

$$\iff T_0 \wedge ((p \vee q) \vee r)$$

$$\iff p \vee q \vee r \iff \text{RHS}$$

—— De Morgan's

—— Distributive

—— Inverse

—— Identity

NAND & NOR CONNECTIVES

The NAND and NOR connectives are represented by $\uparrow$ and $\downarrow$ respectively.

NAND Refers to negation of AND connective

NOR Refers to negation of OR connective.

For any two propositions p & q we have,

$$p \uparrow q \iff \neg(p \wedge q) \iff \neg p \vee \neg q$$

$$p \downarrow q \iff \neg(p \vee q) \iff \neg p \wedge \neg q$$

* Construct truth table for $p \uparrow q$ and $p \downarrow q$. and infer the thumb rule for the same.

For NAND,

Thumb rule: If both p & q are true then $p \uparrow q$ is false.

p	q	$p \wedge q$	$p \uparrow q$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

For NOR,

Thumb Rule: If both p & q are false then $p \downarrow q$ is true.

p	q	$p \vee q$	$p \downarrow q$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

Q) using truth table show that the following is logically equivalent.

$$1) \neg(p \uparrow q) \iff (\neg p \downarrow \neg q)$$

P	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee \neg q$	$\neg(p \uparrow q)$	$(\neg p \downarrow \neg q)$
0	0	1	1	0	1	0	0
0	1	1	0	0	1	0	0
1	0	0	1	0	1	0	0
1	1	0	0	1	0	1	1

Converse, Inverse & Contrapositive of Implication

* construct truth table for the following

$$p \rightarrow q, q \rightarrow p, \neg p \rightarrow \neg q, \neg q \rightarrow \neg p$$

P	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
0	0	1	1	1	1	1	1
0	1	1	0	1 (1)	0 (2)	0 (3)	1 (4)
1	0	0	1	0	1	1	0
1	1	0	0	1	1	1	1

NOTE :- 1) Since column 1 & 4 have same truth values. we can say that they are logically equivalent i.e. $p \rightarrow q \iff \neg q \rightarrow \neg p$

1) we observe that column 2 & 3 have same truth values. therefore, we note that $q \rightarrow p \iff \neg p \rightarrow \neg q$

2) If $p \rightarrow q$ is the implication given, we have the Converse, inverse and contrapositive of the implication.

(i) converse of $p \rightarrow q$ is denoted as $q \rightarrow p$.

(ii) Inverse of $p \rightarrow q$ is denoted as $\neg p \rightarrow \neg q$.

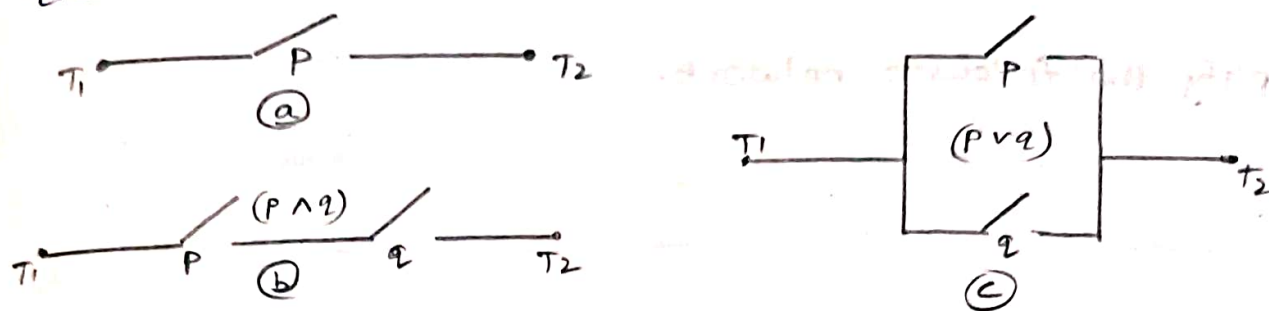
(iii) Contrapositive of $p \rightarrow q$ is denoted as $\neg q \rightarrow \neg p$.

3) i) Implication is logically equivalent to its contrapositive.

ii) Converse of implication is logically equivalent to the inverse.

Switching Networks

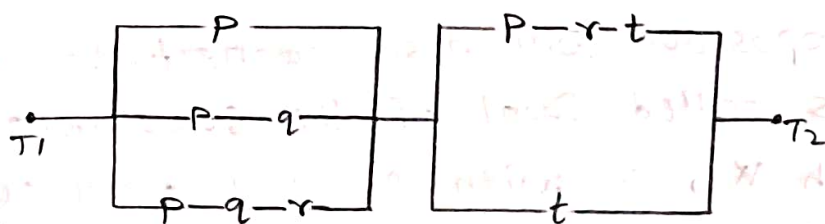
We can construct networks made up of wires and switches connecting to the terminals T_1 & T_2 , where each switch is either open (0/False) so that no current flows through it or closed (1/True) so that current flows through it. The below figures represent the networks with 1 switch and 2 switches respectively.



In figure (b), represents a serial network wherein the current flows from T_1 to T_2 when both the switches P & Q are closed. Such a network can be represented with AND connective represented as $P \wedge Q$.

In figure (c), represents a parallel network wherein the current from T_1 to T_2 when any one of the switches P & Q are closed. Such a network can be represented with OR connective represented as $P \vee Q$.

Q) Simplify the following network.



Sol The above Switching network can be represented equivalently in the proposition form using AND & OR connectives and the propositions P, Q, R, T as given below.

$$[P \vee (P \wedge Q) \vee (P \wedge Q \wedge R)] \wedge [(P \wedge R \wedge T) \vee T]$$

Simplifying the above compound proposition using laws of logic.

$$= [p \vee (p \wedge q) \vee (p \wedge q \wedge r)] \wedge [(p \wedge r \wedge t) \vee t]$$

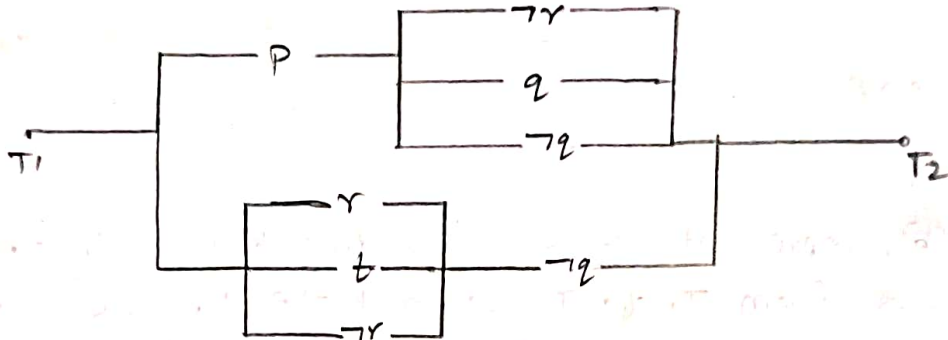
$$= [p \vee (p \wedge q \wedge r)] \wedge [(p \wedge r) \vee t]$$

$$= [p \vee (p \wedge (q \wedge r))] \wedge [t] = (p \wedge t)$$

The equivalent simplified switching network is given below.



Q) Simplify the following network.



Sol:- $[p \wedge (\neg r \vee q \vee \neg q)] \vee [(r \vee t \vee \neg r) \wedge \neg q]$

$$= [p \wedge T_0] \vee [T_0 \wedge \neg q]$$

$$= p \vee \bar{q} = p \vee \neg q$$

Dual / Duality of a Proposition

If 's' is a proposition with basic connectivities $\wedge, \vee$ only then s^d is called Dual of the statement by replacing $\wedge$ with $\vee$, $\vee$ with $\wedge$ and every occurrence of T_0 & F_0 with F_0 and T_0 respectively.

ex Consider $S = (p \wedge \neg q) \vee (r \wedge T_0)$

$$S^d = (p \vee \neg q) \wedge (r \vee F_0)$$

Principle of Duality

If s & t are statements containing only ~~AND~~ & OR connectives then the principle states that if $s \Leftrightarrow t$ then $s^d \Leftrightarrow t^d$

Q) Write the dual for the following

① $q \rightarrow p$

Sol $q \rightarrow p \Leftrightarrow \neg q \vee p$

$\therefore S: \neg q \vee p$

$S^d: \neg q \wedge p$

② $p \rightarrow (q \wedge r)$

Sol $\neg p \vee (q \wedge r) \Leftrightarrow p \rightarrow (q \wedge r)$

$\therefore S: \neg p \vee (q \wedge r)$

$S^d: \neg p \wedge (q \vee r)$

③ $\neg p \uparrow (q \vee r)$

Sol $\neg p \uparrow (q \vee r) \Leftrightarrow \neg(\neg p \wedge (q \vee r))$

$\Leftrightarrow \neg \neg p \vee \neg(q \vee r)$

$\Leftrightarrow p \vee (\neg q \wedge \neg r)$

$\therefore S: p \vee (\neg q \wedge \neg r)$

$S^d: p \wedge (\neg q \vee \neg r)$

+ Expressing the propositions in English statements to the respective symbols and vice-versa

1) Conjunction - $\wedge$ is represented by the word "AND", "BUT"

2) Disjunction - $\vee$ is represented by the word "OR", "Either"

3) Implication/conditional statement -

If $p \rightarrow q$ represents the conditional statement then it can be verbally written as

(i) If p then q

(ii) p is sufficient for q

(iii) q is necessary condition for p

(iv) p only if q .

4) Biconditional- If P and Q are primitive statements, then $P \leftrightarrow Q$ can be verbally written as

- (i) P if and only if Q
- (ii) P is necessary and sufficient for Q
- (iii) P iff Q

ex) Let s, t, u represent the following primitive statements

s : Raghu goes out for walk

t : The Moon is out

u : It is snowing.

Convert the following english sentences into logical notation.

(i) Raghu will go out for walking if and only if the moon is out. $s \leftrightarrow t$

(ii) If it is snowing and the moon is out, then raghu will not go out for walk. $(u \wedge t) \rightarrow \neg s$

(iii) It is snowing but, Raghu will still go out for a walk. $u \wedge s$

Convert the following logics to sentences:

i) $(t \wedge \neg u) \rightarrow s$

If the moon is out and it is not snowing then Raghu goes out for a walk.

ii) $t \rightarrow (\neg u \rightarrow s)$

If the moon is out then if it is not snowing then it is sufficient for Raghu to go out for a walk.

iii) $\neg(s \leftrightarrow (u \vee t))$

Raghu goes out for a walk if and only if it is snowing or the moon is out.

Q 1) Let p, q be the primitive statements

p : John goes to lake George

q : Mary pays for John's shopping.

(i) $p \rightarrow q$ & negate the implication and also represent it using appropriate English statements.

NOTE:- Negation of any implication is not any other implication i.e. it should not begin with if. You must convert the implication into its basic form and apply the negation.

Negation of $p \rightarrow q$ is $\neg(p \rightarrow q)$

$$\Leftrightarrow \neg(\neg p \vee q)$$

$$\Leftrightarrow \neg\neg p \wedge \neg q$$

$$\Leftrightarrow p \wedge \neg q$$

$$\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q.$$

↓

John goes to lake George but Mary does not pay for John's shopping.

(ii) Considering the previous p, q statements let r be another statement.

r : Alice receives a birthday gift

Negate the implication $(p \vee q) \rightarrow r$ and express in English sentence.

Negation of $(p \vee q) \rightarrow r$ is $\neg((p \vee q) \rightarrow r)$

$$\Leftrightarrow \neg(\neg(p \vee q) \vee r)$$

$$\Leftrightarrow (p \vee q) \wedge \neg r$$

John goes to lake George or Mary pays for John's shopping but Alice doesn't receive a birthday gift

* Validity, Proving validity by an argument, truth table & Rules of equivalence

Argument:- An argument is made up of an implication, where the hypothesis is a set of statements called premises and a statement called conclusion of the argument. The general form of the argument is

$$\underbrace{P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n}_{\substack{\text{Hypothesis} \\ \text{set of premises}}} \longrightarrow \underbrace{q}_{\text{Conclusion}}$$

Here we need to examine the premises and prove that the conclusion 'q' logically follows from the given set of premises.

If the set of premises $P_1, P_2, \dots, P_n$ are true statements then we can infer (or) deduce that the conclusion is also true. Here the set of premises are conjuncted to form the hypothesis of the argument.

Valid Argument:- In the argument if all premises have the truth value true we need to find the circumstance the conclusion is true i.e. 'q' has a truth value 'true' then argument is also called a valid argument and also the implication is a 'tautology'.

eg) P : Ravi studies

q : Ravi plays Basketball

r : Ravi passes DMS.

Let P_1, P_2, P_3 be the premises

P_1 : If Ravi studies then he will pass DMS.

P_2 : Ravi does not play basketball then he will study

P_3 : Ravi failed in DMS

Determine whether the argument is valid

$$(P_1 \wedge P_2 \wedge P_3) \rightarrow q$$

NOTE:- The validity of the argument is determined by

- 1) Using Truth tables
- 2) Rules of equivalence

Sol. To prove $P_1 \wedge P_2 \wedge P_3 \rightarrow q$, we construct the truth table and observe the rows wherein if P_1, P_2, P_3 are true then q must be true. If satisfies then it is a valid argument or it is not. also show that the above implication is a tautology.

Expressing premises in terms of logical notation.

$$P_1: P \rightarrow r$$

$$P_2: \neg q \rightarrow P$$

$$P_3: \neg r$$

P_1	P_2	P_3	$\neg q$	$\neg r$	$P \rightarrow r$	$\neg q \rightarrow P$	$(P_1 \wedge P_2 \wedge P_3) \rightarrow q$
0	0	0	1	1	1	0	1
0	0	1	1	0	1	0	1
0	1	0	0	1	1	1	1
0	1	1	0	0	1	1	1
1	0	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	1	0	0	1	0	1	1
1	1	1	0	0	1	1	1

We can observe, in the 3rd row of truth table premises are true and also the truth values of conclusion is true. Hence we infer the argument is valid. we also observe that the implication is tautology.

ex (02) Consider the below valid argument. Identify Identify the rows in truth table that are crucial for assuming the validity of argument also show what rows can be ignored.

$$[P \wedge (P \rightarrow r) \wedge r] \rightarrow [(P \vee q) \rightarrow r]$$

Rules of Inference

Rules of Inference	Related Logical Implication	Name
① $\begin{array}{l} P \\ P \rightarrow q \\ \hline \therefore q \end{array}$	$[P \wedge (P \rightarrow q)] \rightarrow q$	Modus Ponens/ Rule of Detachment
② $\begin{array}{l} P \rightarrow q \\ q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array}$	$[(P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (P \rightarrow r)$	Law of syllogism
③ $\begin{array}{l} P \\ q \\ \hline P \wedge q \end{array}$	$P \wedge q \rightarrow (P \wedge q)$	Rule of conjunction
④ $\begin{array}{l} P \rightarrow q \\ \neg q \\ \hline \therefore \neg P \end{array}$	$(P \rightarrow q) \wedge (\neg q) \rightarrow \neg P$	Modus Tollens Rule
⑤ $\begin{array}{l} P \vee q \\ \neg P \\ \hline \therefore q \end{array}$	$(P \vee q) \wedge (\neg P) \rightarrow q$	Rules of disjunctive Syllogism
⑥ $\begin{array}{l} P \wedge q \\ \hline \therefore P \end{array}$	$P \wedge q \rightarrow P$	Rule of conjunctive simplification
⑦ $\begin{array}{l} P \\ \hline \therefore P \vee q \end{array}$	$P \rightarrow P \vee q$	Rule of disjunctive amplification
⑧ $\begin{array}{l} P \wedge q \\ P \rightarrow (q \rightarrow r) \\ \hline \therefore r \end{array}$	$(P \wedge q) \wedge (P \rightarrow (q \rightarrow r))$	Rule of conditional proof
⑨ $\begin{array}{l} P \rightarrow q \\ r \rightarrow s \\ P \vee r \\ \hline \therefore q \vee s \end{array}$	$(P \rightarrow q) \wedge (r \rightarrow s) \wedge (P \vee r) \rightarrow (q \vee s)$	Rule of constructive dilemma

$$\begin{array}{l} \textcircled{9} \quad P \rightarrow Q \\ \quad \quad r \rightarrow s \\ \quad \quad \neg Q \vee \neg s \\ \hline \therefore \neg P \vee \neg r \end{array}$$

Rule of
disjunctive
dilemma.

$$\textcircled{11} \quad \frac{\neg P \rightarrow F_0}{\therefore P}$$

aku4elk

Method of proof
by contradiction

Q) Test the validity of arguments by symbolic representation.

- ① If ravi gets the supervisor position and works hard then he will get a raise. If he gets a raise, he will buy a new car. He hasn't purchase a new car. Therefore, either ravi did not get supervisor position or he did not work hard.

for the given argument, we can represent in symbolic form by using the propositions given below.

P :

Q :

r :

s :

The above arguments can be written in terms of premises and conclusion as below.

$$\begin{array}{l} (P \wedge Q) \rightarrow r \\ \quad \quad r \rightarrow s \\ \quad \quad \neg s \\ \hline \therefore \neg P \vee \neg Q \end{array}$$

Steps

① $(P \wedge Q) \rightarrow r$

② $r \rightarrow s$

③ $\neg s$

④ $P \wedge Q \rightarrow s$

⑤ $\neg (P \wedge Q)$

⑥ $\neg P \vee \neg Q$

Reasons

premises

premises

premises

①, ② Law of syllogism

④, ③ modus tollens's Rule

⑤ Demorgan's law

- (ii) If the band could not play rock music — — —
(solved in Tutorial)

* Prove by Contradiction

To establish the validity of the argument to prove by contradiction we must assume the negation of conclusion as another premise wherein we use all the premises including the assumed premise to derive the contradiction F_0 . If this objective is achieved then the given conclusion logically follows from the premises.

ex Establish the validity of below arguments by prove by contradiction method.

$$\begin{array}{l} \neg p \leftrightarrow q \\ q \rightarrow r \\ \neg r \\ \hline \therefore p \end{array}$$

NOTE:

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

<u>Sl</u>	<u>Steps</u>	<u>Reason</u>
①	$\neg p \leftrightarrow q$	premises
②	$q \rightarrow r$	premises
③	$\neg r$	premises
④	$\neg p$	Assumed premise (prove by Contradict)
⑤	$(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	Logical equivalence $p \leftrightarrow q$
⑥	$\neg q$	②, ③ modus Tollens
⑦	$\neg p \rightarrow q$	conjunctive simplification.
⑧	$\neg \neg p$	⑤, ⑦ modus Tollens
⑨	p	⑧ Double negation law
⑩	$p \wedge \neg p$	④, ⑨ Rule of conjunction
⑪	F_0	⑩ Inverse law
⑫	p	④, ⑪ proof by contradiction

$$\begin{array}{l}
 \text{Q)} \quad p \rightarrow (q \wedge r) \\
 r \rightarrow s \\
 \neg(q \wedge s) \\
 \hline
 \therefore \neg p
 \end{array}$$

$$\neg s \rightarrow \neg r$$

$$\neg q \vee \neg s \rightarrow \neg s \vee \neg q$$

$$s \rightarrow \neg q$$

$$r \rightarrow s$$

Steps

$$\textcircled{1} \quad p \rightarrow (q \wedge r)$$

$$\textcircled{2} \quad r \rightarrow s$$

$$\textcircled{3} \quad \neg(q \wedge s)$$

$$\textcircled{4} \quad p$$

$$\textcircled{5} \quad \neg q \vee \neg s$$

$$\textcircled{6} \quad \neg s \vee \neg q$$

$$\textcircled{7} \quad s \rightarrow \neg q$$

$$\textcircled{8} \quad r \rightarrow \neg q$$

$$\textcircled{9} \quad \neg r \vee \neg q$$

$$\textcircled{10} \quad \neg(q \wedge r)$$

$$\textcircled{11} \quad \neg p$$

$$\textcircled{12} \quad p \wedge \neg p$$

$$\textcircled{13} \quad F.$$

$$\textcircled{14} \quad \neg p$$

Reasons

Premises

Premises

Premises

Assumed premise

③ Demorgans law

⑤ commutative law

⑥ $p \rightarrow q$

②, ⑦ law of syllogism

⑧ $p \rightarrow q$

④ Demorgans law.

①, ⑩ modus Tollens

④, ⑪ Rule of conjunction

⑫ Inverse law

④, ⑬ proof by contradiction

Open Statements

Consider the below statements,

$$x^2 = 2, \quad x + 5 = 8, \quad x + 2 \text{ is an even integer.}$$

These statements are not propositions because the value of 'x' can't be determined unless the variable 'x' is substituted with a value.

Such statements are referred to as open statement. We can restrict our choices of 'x' to be only integers.

Then if x is equal to 3, Then $x^2 = 2$ is false.

$x+5=8$ is evaluated as true and $x+2$ is an even integer is false. A declarative statement is an open statement if and only if it contains 1 or more variables. It is not a proposition. It becomes proposition when variables are replaced by a certain allowable choices referred to as Universe (U).

Open statements are denoted by $P(x), Q(x), r(x,y), \dots$ where x & y are free variables.

$P(x): x+2$ is an even integer

$\neg P(x): x+2$ is an odd integer.

$Q(x,y): x+y$ is a multiple of 3.

if we consider $a \in U$. then, replacing ' x ' by ' a ' we get $P(a)$ where its truth value can be determined.

Q) Consider the following open statements and universe as integers.

$P(x): x \leq 3$

$r(x): x > 0$

$Q(x): x+1$ is odd

(i) $P(2)$

$P(2): 2 \leq 3$

$\therefore P(2)$ - Truth value is true

(iii) $P(2) \wedge Q(3)$

$P(2): 2 \leq 3 \rightarrow \text{True}$

$Q(3): 3+1$ is odd $\rightarrow \text{False}$

True & False $\rightarrow \text{False}$

$P(2) \wedge Q(3)$ - Truth value is false.

(ii) $\neg Q(4)$

$Q(x): x+1$ is odd

$\neg Q(x): x+1$ is even

$\neg Q(4): 4+1$ is odd

$Q(4):$ is even

$Q(4)$ - Truth value is false

(v) $P(1) \leftrightarrow Q(2)$

$P(1): 1 \leq 3 \rightarrow \text{True}$

$Q(2): 2+1$ is odd $\rightarrow \text{True}$

True $\leftrightarrow$ True = True

$P(1) \leftrightarrow Q(2)$ - Truth value is True

(iv) $r(-3) \rightarrow q(5)$

Here, $r(-3) : -3 > 0$ is evaluated as false

$\therefore r(-3) \rightarrow q(5)$ is evaluated as True.

Since, wkt if the hypothesis truth value is false the respective of the truth value of the implication becomes True. $\therefore r(-3) \rightarrow q(5)$ is true.

* Quantifiers:

The (quantifiers) open statements can be quantified by using 2 types of quantifiers.

- 1) Universal quantifiers - refers to all the elements in the universe
- 2) Existential quantifiers - refer to atleast one element in the universe.

phrases used

Existential

- for some 'x'
- for atleast 1 'x'
- There exist 'x'
- symbolically written as $\exists$

Universal

- for all 'x'
- for any 'x'
- for each 'x'
- for every 'x'
- symbolically represented as $\forall$

In quantifiers statements the variables are known as bound variables. Quantifiers can be used along with logical connectivities, implication, we can have logical equivalence and also apply laws of logic and rules of Inference.

Q) Determine the truth value for the below quantifier statements by considering the universe real numbers and the below open statement

$$P(x): x \geq 0$$

$$r(x): x^2 - 3x - 4 \geq 0$$

$$q(x): x^2 \geq 0$$

$$s(x): x^2 - 3 \geq 0$$

① $\exists x (P(x) \wedge q(x))$

prove that its truth value is true

consider the value of x as 6 which belongs to the universe of real numbers.

$$\therefore P(6): 6 \geq 0 \text{ is true}$$

$$q(6): 36 \geq 0 \text{ is true}$$

$\therefore$ we conclude that $\exists$ (there exist) x $\exists x (P(x) \wedge q(x))$ is true by considering $x = 6$.

② $\forall x (P(x) \rightarrow q(x))$

prove that its truth value is true

for all +ve, -ve integers
the condition is satisfied

③ $\forall x (q(x) \rightarrow s(x))$

prove that its truth value is false

put $x = 0$, condition is failed

④ $\forall x (r(x) \rightarrow p(x))$

prove that its truth value is false

put $x = -1$,

condition is failed.

Q) Let $p(x)$, $q(x)$ & $r(x)$ be the following open statements.

$$p(x) : x^2 - 7x + 10 = 0 \quad 5, 2$$

$$q(x) : x^2 - 2x - 3 = 0 \quad 3, -1$$

$$r(x) : x < 0$$

$$\textcircled{1} \forall x (p(x) \rightarrow q(x))$$

$$\text{at } x = 5, x = 2$$

the condition fails.

$$\textcircled{2} \forall x (q(x) \rightarrow r(x))$$

$$\text{at } x = 3$$

the condition fails

$$\textcircled{3} \exists x (q(x) \rightarrow r(x))$$

$$\text{at } x = -1$$

the condition satisfies

$$\textcircled{4} \exists x (p(x) \rightarrow r(x))$$

$$\text{at } x = 5 \text{ (or) } x = 2$$

the condition fails.

* Rules for negation of quantified statements with one variable

$$\textcircled{1} \neg(\forall x(P(x))) \iff \exists x(\neg P(x))$$

$$\textcircled{2} \neg(\exists x(P(x))) \iff \forall x(\neg P(x))$$

$$\textcircled{3} \neg(\forall x(\neg P(x))) \iff \exists x(\neg\neg P(x)) \iff \exists x(P(x))$$

$$\textcircled{4} \neg(\exists x\neg P(x)) \iff x(\neg\neg P(x)) \iff \forall x(P(x))$$

* Logical equivalence for quantified statements

$$\textcircled{1} \forall x \neg(P(x) \wedge Q(x)) \iff \forall x (\neg P(x) \vee \neg Q(x))$$

$$\textcircled{2} \forall x \neg(P(x) \vee Q(x)) \iff \forall x (\neg P(x) \wedge \neg Q(x))$$

$$\textcircled{3} \forall x \neg\neg P(x) \iff \forall x P(x)$$

$$\textcircled{4} \exists x (P(x) \rightarrow Q(x)) \iff \exists x (\neg P(x) \vee Q(x))$$

$$\textcircled{5} \forall x (P(x) \wedge Q(x)) \iff \forall x P(x) \wedge \forall x Q(x)$$

$$\textcircled{6} \exists x (P(x) \vee Q(x)) \iff \exists x (P(x)) \vee \exists x (Q(x))$$

Q) Negate and Simplify the following.

$$(i) \exists x [P(x) \vee Q(x)]$$

$$\text{Negation} \rightarrow \neg \exists x (P(x) \vee Q(x))$$

$$\iff \forall x \neg (P(x) \vee Q(x)) \quad \text{Negation rule}$$

$$\iff \forall x (\neg P(x) \wedge \neg Q(x)) \quad \text{De Morgan's law}$$

$$(ii) \forall x (P(x) \wedge \neg Q(x))$$

$$\text{Negation} \rightarrow \neg \forall x (P(x) \wedge \neg Q(x))$$

$$\iff \exists x \neg (P(x) \wedge \neg Q(x))$$

$$\iff \exists x (\neg P(x) \vee Q(x))$$

$$(iii) \forall x (P(x) \rightarrow Q(x))$$

$$\text{Negation} \rightarrow \neg \forall x (P(x) \rightarrow Q(x))$$

$$\iff \neg \forall x (\neg P(x) \vee Q(x)) \iff \exists x \neg (\neg P(x) \vee Q(x))$$

$$\iff \exists x (P(x) \wedge \neg Q(x))$$

$$(iv) \exists x ((p(x) \vee q(x)) \rightarrow r(x))$$

$$\text{Negation} \rightarrow \neg \exists x ((p(x) \vee q(x)) \rightarrow r(x))$$

$$\Leftrightarrow \neg \exists x (\neg (p(x) \vee q(x)) \vee r(x))$$

$$\Leftrightarrow \neg \exists x (\neg p(x) \wedge \neg q(x) \vee r(x))$$

$$\Leftrightarrow \forall x \neg (\neg p(x) \wedge \neg q(x) \vee r(x))$$

$$\Leftrightarrow \forall x (p(x) \vee q(x) \wedge \neg r(x))$$

* Rule of Universal Specification

If an open statement is true for all the replacement by the members in the given universe then, the open statement is True for a specific member in the Universe.

If $p(x)$ is an open statement and if $\forall x$, $p(x)$ is True for $x \in U$. Therefore, $p(c)$ is True where 'c' is some arbitrary element belonging to the universe.

* Rule of Universal Generalization

If $p(x)$ is an open statement, If $p(c)$ is True for all values of $c \in U$, then $\forall x p(x)$ is True.

NOTE:- The above two rules are applied for existential quantifiers.

- Rule of existential specification

$$\exists x p(x) \text{ is True}$$

$$\therefore p(x) \text{ is True, where } c \in U$$

- Rule of existential generalization

$$\text{If } p(c) \text{ is true, where } c \in U$$

$$\therefore \exists x \text{ is true.}$$

Q) Establish the validity of following arguments.

$$\begin{array}{l} (i) \quad \forall x [P(x) \rightarrow Q(x)] \\ \quad \forall x [Q(x) \rightarrow R(x)] \\ \hline \therefore \forall x [P(x) \rightarrow R(x)] \end{array}$$

Steps

$$\textcircled{1} \quad \forall x [P(x) \rightarrow Q(x)]$$

$$\textcircled{2} \quad \forall x [Q(x) \rightarrow R(x)]$$

$$\textcircled{3} \quad P(c) \rightarrow Q(c)$$

$$\textcircled{4} \quad Q(c) \rightarrow R(c)$$

$$\textcircled{5} \quad P(c) \rightarrow R(c)$$

$$\textcircled{6} \quad \forall x [P(x) \rightarrow R(x)]$$

Reasons

Premises

Premises

$\textcircled{1}$, Rule of universal specification
consider $P(c)$ on an arbitrary element

$\textcircled{2}$, Rule of universal specification
consider $Q(c)$ on an arbitrary element

$\textcircled{3}\textcircled{4}$ applying law of syllogism

$\textcircled{5}$ Rule of universal generalization

(ii) For the universe of all people, consider the below open statements.

$m(x)$: x is a Mathematics Professor

$c(x)$: x has studied Calculus.

Consider below arguments and prove its validity.

a) All mathematics professors studied Calculus

b) Mary is mathematics professor

c) Mary studied Calculus.

Ans considering the above arguments, Mary as a particular person in the universe, represented as 'a'.
The above argument can be written in symbolic form as below.

$$\forall x (m(x) \rightarrow c(x))$$

$$m(a)$$

$$\therefore c(a)$$

→ Definitions & Theorems: Till now the argument has proved its validity by making use of laws of logic, Rules of inference, RUG & RUG and This approach is too difficult job to achieve proving certain theorems in mathematics. To assist this there are 3 ways of proving "direct method" and "indirect method" and "proof by contradiction".

• Direct proof: - To prove that $P \rightarrow Q$ is a valid argument

i) Hypothesis: Assume 'P' is true.

ii) Analysis: Using the definitions and rules infer that 'Q' is true.

iii) Conclusion: Therefore, $P \rightarrow Q$ is true.

• Indirect proof: - To prove $P \rightarrow Q$, we apply direct proof to its contrapositive i.e. $\neg Q \rightarrow \neg P$

i) Hypothesis: Assume $\neg Q$ is true.

ii) Analysis: Using the definitions and rules infer that ' $\neg P$ ' is true.

iii) Conclusion: Therefore, $P \rightarrow Q$ is true.

• Proof by contradiction: -

i) Hypothesis: Assume $P \rightarrow Q$ is false.

ii) Analysis: Using the definitions and rules infer that 'P' is false, this contradicts the assumption that 'P' is true.

iii) Conclusion: Since, the contradiction is arrived because of false assumption made. Therefore, we infer that $P \rightarrow Q$ is true.

Definitions:- Let 'n' be an integer, we call 'n' is even when it is a multiple of 2 i.e. there exist an integer "a" such that $n = 2a$ where $a \in \mathbb{Z}$

lly, If 'n' is odd then 'n' can be expressed as $n = 2a + 1$, $a \in \mathbb{Z}$

Theorems

① If 'm' is an even integer then 'm+7' is odd

Prove the above theorem by direct method & indirect method.

Proof:- Step-1:- Hypothesis:- Assume that m is an even integer $\therefore m = 2a$, $a \in \mathbb{Z}$

Direct method

Step-2:- Analysis:- Using the definition infer that m+7 is odd

we have, $m = 2a$.

$$m+7 = 2a+7$$

$$m+7 = 2a+6+1$$

$$m+7 = 2(a+3)+1$$

$$\therefore m+7 = 2c+1 \text{ where } c = a+3, c \in \mathbb{Z}$$

Step-3: Conclusion: we conclude that if m is even integer then m+7 is odd integer.

Proof:- Step-1:- Hypothesis: Assume that m+7 is even integer.

Indirect method

$$\therefore m+7 = 2a, a \in \mathbb{Z}$$

Step-2:- Analysis: using definition, we must infer that m is odd.

$$\text{we have } m+7 = 2a$$

$$m = 2a - 7 \Rightarrow m = 2a - 8 + 1$$

$$= 2(a-4) + 1$$

$$\therefore m = 2c + 1 \text{ where } c = a - 1$$

$\therefore m$ is odd

Step-3:- Conclusion:- We conclude that if $m+7$ is even then m is odd hence the above statement is valid.

② For all integers if k & l are both odd then $k+l$ is even.

Give direct proof for this statement.

Proof Step-1:- Hypothesis:- Assume that k & l both are odd integers

$$\therefore k = 2a + 1, \quad l = 2b + 1 \text{ where } a, b \in \mathbb{Z}$$

Step-2:- Analysis:- Using the definition infer that $k+l$ is even.

$$\text{we have, } k = 2a + 1$$

$$l = 2b + 1$$

$$k+l = 2a + 2b + 1 + 1 = 2a + 2b + 2$$

$$= 2(a + b + 1)$$

$$= 2c \text{ where } c = a + b + 1, \quad c \in \mathbb{Z}$$

Step-3:- Conclusion:- We conclude that if k, l are both odd then $k+l$ is even.

② Give direct, indirect, contradiction proof for the theorem "If n is an odd integer then n^2 is odd".

Proof Step-1:- Hypothesis:- Assume that n is an odd integer
Direct method
 $n = 2a + 1, \quad a \in \mathbb{Z}$

Step-2:- Analysis:- Using definition and rules infer that

$$c = 2a^2 + 2a$$

$$c \in \mathbb{Z}$$

$$n^2 = n \times n = (2a+1)(2a+1)$$

$$= 4a^2 + 4a + 1$$

$$= 2(2a^2 + 2a) + 1$$

$$n^2 = 2c + 1$$

Step-3: Conclusion: we conclude that when n is odd then n^2 is also odd.

Proof: Step-1: Hypothesis: Assume n^2 is even integer
Indirect Method
 $n^2 = 4a^2, n \in \mathbb{Z}$

Step-2: Analysis: Using definition and theorems we infer that n is even.

$$n^2 = 4a^2 = 2a \times 2a$$

$$\therefore n \in \mathbb{Z}.$$

$$\Rightarrow n = 2a$$

$\therefore n$ is even.

Step-3: Conclusion: we conclude that when n^2 is even then n is also an even integer.

Proof:
Contradiction

NOTE:- We have a valid argument for any two statements P and Q where $P \rightarrow Q$ is a tautology. This is represented by $P \Rightarrow Q$ which in other words said as " P logically implies Q " (or) " P logically follows Q ".

Q) Show that, $P \rightarrow Q, \neg r$ are logically implies negation P .

Sol $[(P \rightarrow Q) \wedge (\neg r)] \Rightarrow \neg P$

The above problem is symbolically written as $P \rightarrow Q, \neg r$ logically implies $\neg P$ as above.

To verify it is a logical implication, we must check whether $[(P \rightarrow Q) \wedge (\neg r)] \rightarrow \neg P$ is a tautology.

To prove this we construct the truth table as follows.

P	q	r	$P \rightarrow q$	$\neg r$	$(P \rightarrow q) \wedge \neg r$	$\neg P$	$[(P \rightarrow q) \wedge \neg r] \rightarrow \neg P$
0	0	0	1	1	1	1	1
0	0	1	1	0	0	1	1
0	1	0	1	1	1	1	1
0	1	1	1	0	0	1	1
1	0	0	0	1	0	0	1
1	0	1	0	0	0	0	1
1	1	0	1	1	1	0	0
1	1	1	1	0	0	0	1

From the last column, we can observe that
It is not a tautology. Hence, we conclude that
 $P \rightarrow q$, $\neg r$ does not logically implies / follows $\neg P$.

Q) Establish the validity of the below argument.

$$\forall x [P(x) \vee q(x)]$$

$$\forall x [(\neg P(x) \wedge q(x)) \rightarrow r(x)]$$

$$\therefore \forall x [\neg r(x) \rightarrow P(x)]$$

HINT:- Assume $\neg r(x)$ as one of the premise and
arrive at the conclusion: $p(c)$ for some element
'c' in the universe. Hence conclude that $\neg r(c) \rightarrow p(c)$.
following $\forall x [\neg r(x) \rightarrow p(x)]$

- Sol
- | | |
|--|-----------------------|
| ① $\forall x [P(x) \vee q(x)]$ | Premises |
| ② $\forall x [(\neg P(x) \wedge q(x)) \rightarrow r(x)]$ | Premises |
| ③ $\neg r(c)$ | Assumed premise |
| ④ $P(c) \vee q(c)$ | RUS |
| ⑤ $(\neg P(c) \wedge q(c)) \rightarrow r(c)$ | RUS |
| ⑥ $\neg (\neg P(c) \wedge q(c))$ | ③⑤ Modus Tollens |
| ⑦ $\neg \neg P(c) \vee \neg q(c)$ | Demorgan's |
| ⑧ $P(c) \vee \neg q(c)$ | Double negation |
| ⑨ $P(c) \vee (q(c) \wedge \neg q(c))$ | ⑧, ⑦ Distributive law |

(10) $\neg \neg F_0$

Inverse law

(11) $P(c)$

Identity law

(12) $\neg Y(c) \rightarrow P(c)$

(13) $\forall x (\neg Y(x) \rightarrow P(x))$

RUG.